What is life?

Characteristics of living things

Most of us can recognise the difference between living and non-living things. Living organisms are identified by a number of characteristics. They:

- move, or are in some way active (eg sessile (non-moving) organisms actively take up nutrients)
- have a recognisable size and shape (per species)
- respond to, and play an active role in, their external environment
- take up nutrients (organic compounds) from the environment
- reproduce ('life arises from life')
- have a reasonably predictable life span (eg die at a predictable 'old age' barring accident or early onset of disease)

'Life was never meant to be easy' is a very common statement. Living organisms must fight to survive. They do this using a number of means, including variation and adaptation.

1 Name five sessile organisms.

2 Discuss the strategies organisms use to survive. As well as mentioning 'variation' and 'adaptation', include another survival strategy.

Unicellular and multicellular organisms

Organisms, or their component parts, can be made up of a single cell or many cells. Single-celled organisms are called *unicellular* (*uni* one) while those composed of more than one cell are *multicellular* organisms (*multi* many). With so many cells involved, the cells of multicellular organisms have evolved to specialise, resulting in a division of labour. For example, sperm and eggs are specialised cells that are only involved in reproduction.

Cells with a common embryonic origin (in other words, similar cell types) can group together to share a common structure and function. This structure is called the *tissue* in an organism. Two or more types of tissue can work together to create a functional unit called an *organ*. Organs perform specific tasks, for example the heart acts as a pump to circulate blood. When two or more organs work together an *organ system* develops. The digestive system is an example of an organ system. Organ systems form an organism.

3 Identify four organs that are part of the digestive system.

4 Identify four organs that are part of the circulatory system.

Living or non-living?

There has been much debate as to whether certain things should be classified as living or non-living. Bacteria, fungi, viruses and prions have all been placed into both categories at various times. Bacteria and fungi are classified as living things but viruses and prions are generally regarded as non-living.

Viruses

Viruses have often been called 'the living dead', surrounding them with a science fiction-like mystery. Viruses are little more than units carrying genetic material (DNA or RNA) housed in a mechanism designed to insert it into a living cell. Without the cell, viruses are considered dormant, inactive. But given access to a living cell they attach to it and inject their genetic material into the cell. The cell is then taken over and used to replicate numerous copies of the virus. Eventually, so many are in the cell that it bursts open (lysis) and the flood of newly manufactured viruses repeat the procedure. Without access to a living cell, they remain inactive (for up to thousands of years, depending on the virus). However, once they have access to the cell's organelles, a necessary requirement to replication, they actively reproduce.

5 List five characteristics of a living thing.

6 Draw up a list that summarises the points for and against viruses being classified as living.

7 Give the full name for the acronym *DNA*.

8 Bacteria is the plural name for a single bacterium. What is the singular name for *fungi*?

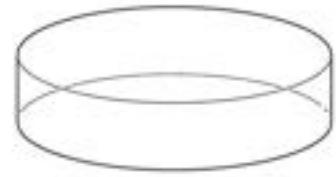
© Wendy St Germain, Peter Gribben and Learning Essentials 2003.

© Blake Education ISBN 978 1 92561 068 0 Assignments in Biology Book 3

Culturing bacteria

In classification, bacteria make up the kingdom known as *Monera*. There are thousands of different species of bacteria and they exist in many different types of environment. Sometimes scientists need to know if bacteria are present in a particular place and, if so, to identify them.

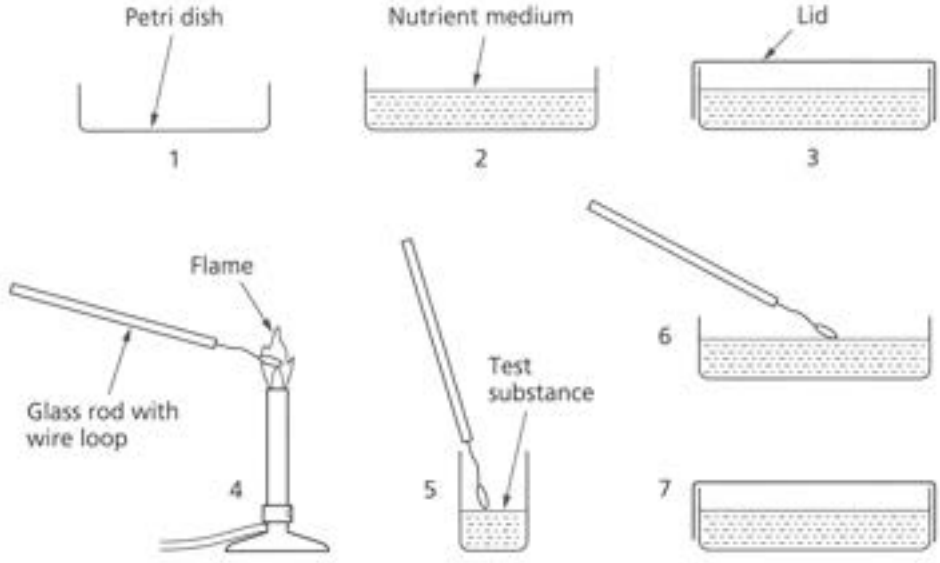
One technique that microbiologists use to grow bacteria (and other microorganisms) is *culturing*. A Petri dish (and lid) is a common piece of apparatus used in culturing bacteria. It is a shallow, circular dish with a cover and was invented in 1887 by R J Petri, a colleague of Robert Koch (1843–1910)*. Petri dishes are made from glass or clear plastic. Microorganisms are grown on a nutrient medium of agar gel. Agar-Agar is obtained from seaweed and contains a powerful gelatinising agent.



Petri dish

* Some sources state that Petri modified Koch's original design.

9 The diagrams show the sequence followed in culturing microorganisms. Describe how this process is carried out, referring to the diagrams.



10 Define *nutrient*.

11 Define *gel*.

12 Define *environment*.

13 Name the types of organisms that make up the kingdom *Monera*.

Decide whether the following statements are true or false:

14 Bacteria replicate by cell division. _____

15 Like viruses, bacteria are not considered to be true organisms. _____

16 Bacteria are unicellular organisms. _____

17 Bacteria are classified on the basis of their shape. _____

Research

18 *Legionnaires' disease* is an illness that was first reported in 1976. Describe the symptoms, name the causal agent, most common location of outbreaks of the disease, and outline the treatment for the disease.

19 Outline some of Robert Koch's contributions to Medicine.

© Wendy St Germain, Peter Gribben and Learning Essentials 2003.

© Blake Education ISBN 978 1 92561 068 0 Assignments in Biology Book 3